HW5 (due Feb. 18, 2010)

1

40 pts.

A paramagnet in one dimension can be modelled as a linear chain of $N+1$ spins. Each spin interacts with its neighbours in such a way that the energy is $U = n\epsilon$ where n is the number of domain walls separating regions of up spins from down as shown by a vertical line in the representation below.

$$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow|\downarrow\downarrow\downarrow\downarrow|\uparrow\uparrow\uparrow\uparrow|\downarrow\downarrow|\uparrow\uparrow\uparrow|\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow$$

How many ways can n domain walls be arranged? Calculate the entropy, $S(U)$, and hence show that the energy is related to the temperature as

$$U = \frac{N\epsilon}{\exp(\epsilon/k_{\mathrm{B}}T) + 1}.$$

Sketch the energy and the heat capacity as a function of temperature, paying particular attention to the asymptotic behaviour for low and for high temperatures.

2

20 pts.

The entropy of black body radiation is given by the formula $S = \frac{4}{3}\sigma V^{1/4}U^{3/4}$ where σ is a constant. Determine the temperature of the radiation and show that $PV = U/3$.

Note: $U = E$ here.

3

40 pts.

Consider a system of N non-interacting particles, each fixed in position and carrying a magnetic moment. The system is immersed in a magnetic induction field $\mathbf{B}$. Each particle may then exist in one of the two energy states: a spin-down state of energy $-\mu B$ and a spin-up state of energy μB. Treat the particles as being distinguishable.

Derive a formula for the entropy in terms of the number of particles in the upper state, n. Sketch $S(n)$. Find the value of n for which $S(n)$ is a maximum. What is the relation between S and the temperature? By treating the energy as a continuous variable, show that the system can have negative temperatures. If a system with a negative temperature makes thermal contact with one with a positive temperature, which way will the heat flow?